

Kardar-Parisi-Zhang universality class in the synchronization of oscillator lattices with time-dependent noise

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Systems of oscillators subject to time-dependent noise typically achieve synchronization for long times when their mutual coupling is sufficiently strong. The dynamical process whereby synchronization is reached can be thought of as a growth process in which an interface formed by the local phase field gradually roughens and eventually saturates. Such a process is here shown to display the generic scale invariance of the one-dimensional Kardar-Parisi-Zhang universality class, including a Tracy-Widom probability distribution for phase fluctuations around their mean. This is revealed by numerical explorations of a variety of oscillator systems: rings of generic phase oscillators and rings of paradigmatic limit-cycle oscillators, like Stuart-Landau and van der Pol. It also agrees with analytical expectations derived under conditions of strong mutual coupling. The nonequilibrium critical behavior that we find is robust and transcends the details of the oscillators considered. Hence, it may well be accessible to experimental ensembles of oscillators in the presence of, e.g., thermal noise.

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Introduction. Critical dynamics, i.e., the time evolution of a spatially extended system at a critical point, is paramount throughout physics [1], biology [2], and society [3]. Recent examples range from active matter [4,5] to driven-dissipative Bose condensates [6,7], quantum magnetism [8,9], or Anderson localization in disordered conductors [10]. One key player in this context is the stochastic Kardar-Parisi-Zhang (KPZ) equation with time-dependent noise [11], which actually displays generic scale invariance (GSI) regardless of parameter values [1,12]. Having been originally put forward as a versatile model [13–15] for the growth of a rough interface or, via suitable mappings, directed polymers in random media, or randomly stirred fluids, this system represents a widely encompassing universality class [16] that transcends, e.g., the classic models A or B of critical dynamics [1] far from equilibrium. Beyond some of the previous examples [6–8,10], the KPZ class further includes instances of, e.g., turbulent liquid crystals [17], thin film growth [18], random geometry [19], active fluids [20], or quantum entanglement [21]. A remarkable trait of the one-dimensional (1D) KPZ universality class is that the fluctuation statistics is described by the Tracy-Widom (TW) family of probability distribution functions (PDF) [16,22]. Indeed, these PDFs have recently been found in systems across different length scales irrespective of their physical nature [23], providing analogs of the Gaussian distribution for strongly correlated variables [24].

A seemingly different, albeit no less paradigmatic, phenomenon is the synchronization of assemblies of coupled oscillators [25–27]. Indeed, this dynamical process bears crucial importance for the collective performance of many systems throughout science and technology from coupled chaotic lasers to swarms of fireflies, spiking neurons, or pacemaker cells in the heart [25–30]. Traditionally, the time evolution of an oscillator array into its synchronized state has

been implicitly assumed to be system specific and nonuniversal. Nevertheless, connections between synchronization and the KPZ equation have been noted along the years [25,31,32]. They have been underscored more recently due to, e.g., the relation between synchronization models and the so-called compact KPZ equation (see, e.g., Ref. [33] and others therein), relevant to quantum systems like exciton-polariton condensates [7]. However, only in the case of idealized oscillator models has the synchronous dynamics been partially shown to display 1D KPZ universal properties or related features [34–36]. If the natural frequencies of the oscillators are disordered and quenched, both the scaling ansatz and the scaling exponent values characterizing the critical dynamics of the synchronization process have been shown to actually differ from 1D KPZ, agreeing, rather, with those of the KPZ equation with columnar disorder [37,38] for 1D lattices of both idealized phase oscillators [39] and of more realistic limit cycle oscillators [40]. Nevertheless, fluctuations are TW distributed in all cases [39,40]. Yet, elucidating the full relation of synchronization in the presence of time-dependent noise to 1D KPZ remains crucial, as such noise occurs in many experimental systems where synchronization emerges [29,41].

In this paper, we show that the space-time process whereby oscillators readjust their frequencies and eventually synchronize displays, for 1D systems with time-dependent noise, the full set of universal properties of the 1D KPZ universality class, i.e., the scaling ansatz, the critical exponent values, and the TW statistics. This is borne out by an extensive numerical analysis of large systems of phase oscillators and of two paradigmatic limit-cycle oscillators, namely, those of Stuart-Landau (SL) and van der Pol (vdP). It is only for odd couplings among the phases (a symmetry unlikely to hold exactly in the phase-reduced dynamics of generic limit-cycle oscillators, yet present in the celebrated Kuramoto model [29,31])

that the critical behavior differs, as it belongs to the Edwards-Wilkinson (EW) universality class, i.e., that of the linearized KPZ equation [13–15]. The robustness of our results against changes in the nature of the oscillators suggests a potential relevance for experiments, and also that of synchronization to critical dynamics at large.

Models. We consider the synchronization of rings of three different self-sustained oscillators in the presence of time-dependent noise. In all cases, the dynamics reads

$$\dot{\mathbf{r}}_j = f(\mathbf{r}_j) + K\Gamma(\mathbf{r}_{j+1} - \mathbf{r}_j) + K\Gamma(\mathbf{r}_{j-1} - \mathbf{r}_j) + \xi_j \quad (1)$$

for $j = 1, 2, \dots, L$, with periodic boundary conditions (PBCs), i.e., $\mathbf{r}_{L+1} \equiv \mathbf{r}_1$ and $\mathbf{r}_0 \equiv \mathbf{r}_L$, where the state vector $\mathbf{r}_j \in \mathbb{R}^n$ (here, $n = 1$ or 2) and dot is time derivative. The autonomous evolution of each dynamical unit is given by the vector field f (see below) and is always chosen so it presents a stable limit cycle with period of oscillation 2π (unit angular frequency). The coupling is diffusive [26] and given by an overall coupling strength $K \geq 0$ and a coupling function Γ , whose range is $\mathcal{O}(1)$. Some of the components $\xi_j^{(a)}$, $a = 1, 2, \dots, n$, of the noise terms may be set to zero in some particular system. The nonzero noise components are independent and Gaussian, with zero mean and variance D , and delta correlated in time, $\langle \xi_j^{(a)}(t) \xi_k^{(b)}(t') \rangle = D \delta_{jk} \delta_{ab} \delta(t - t')$. Our simulations employ the standard Euler-Maruyama scheme [42] with 0.01 time step and are shown for $K = 1$ and $\sqrt{D} = 0.1$.

The first dynamical system that we consider is the Kuramoto-Sakaguchi (KS) phase oscillator [43]. The state variable $\mathbf{r}_j$ (phase) is a scalar, henceforth denoted ϕ_j , while $f(\phi_j) = 1$. The coupling function is $\Gamma_{\text{KS}}(\Delta\phi) = \sin(\Delta\phi + \delta)$, where $\Delta\phi$ is the phase difference between two neighboring oscillators and the parameter $\delta \in \mathbb{R}$. There is a single noise component ($n = 1$).

Our second dynamical system is the SL oscillator. This limit-cycle oscillator provides the universal amplitude equation at a supercritical Hopf bifurcation [44]. The amplitude has two components, described through $z_j \in \mathbb{C}$. The autonomous dynamics, $f(z_j) = (\mu + i)z_j(t) - |z_j(t)|^2 z_j(t)$, has an attracting, circular limit cycle with radius $\sqrt{\mu}$, for $\mu > 0$, with $\mu \in \mathbb{R}$ measuring distance to bifurcation. The coupling function is $\Gamma_{\text{SL}}(\Delta z) = c \Delta z$, where Δz is the state difference between two neighboring oscillators and $c = (1 + i\gamma)/\sqrt{1 + \gamma^2} \in \mathbb{C}$ depends on parameter $\gamma \in \mathbb{R}$. Two additive noise components affect $\text{Re } z(t)$ and $\text{Im } z(t)$.

We finally consider the vdP oscillator, a full-fledged limit-cycle oscillator of experimental relevance, introduced in the context of electronic circuits almost a century ago [44]. The autonomous dynamics reads $\ddot{x}_j = (\mu - x_j^2)\dot{x}_j - x_j$, which, by defining $\mathbf{r}_j = (x_j, v_j \equiv \dot{x}_j)$, implies $f(x_j, v_j) = (v_j, (\mu - x_j^2)v_j - x_j)$. In the absence of coupling and noise, a stable limit cycle ensues with $\mathcal{O}(\sqrt{\mu})$ size for $\mu > 0$, above a supercritical Hopf bifurcation at $\mu = 0$ [44]. The coupling function is $\Gamma_{\text{vdP}}(\Delta x, \Delta v) = (0, (\gamma \Delta x + \Delta v)/\sqrt{1 + \gamma^2})$, where $\Delta x, \Delta v$ are the components of the state difference between two neighboring oscillators, and γ plays a role analogous to the homonymous SL parameter. Noise only enters the equation giving $\dot{v}_j$. For SL and vdP oscillators, we inspect $\mu = 0.5$

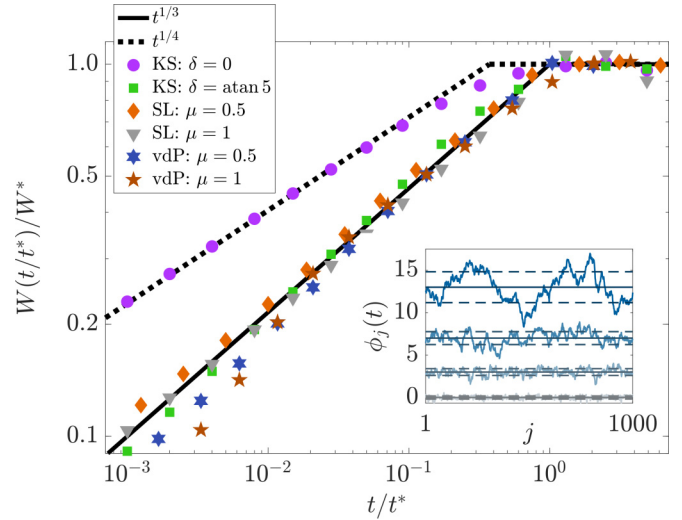


FIG. 1. Roughness of phase profiles in rings of self-sustained oscillators. Main panel: Roughness (2) normalized by its saturation value, W/W^* , as a function of time rescaled by the saturation time, t/t^* , for rings of $L = 1000$ oscillators of all three dynamical systems under consideration (see legend). While the KS rings with $\delta = 0$ display growth exponent $\beta = \beta_{\text{EW}} = 1/4$, all other rings of oscillators display $\beta = \beta_{\text{KPZ}} = 1/3$. Five-thousand noise realizations have been used in the averaging. Inset: Phases $\phi_j(t)$ in a ring of $L = 1000$ KS oscillators with $\delta = \text{atan}5$, forming a phase profile that becomes progressively rougher, at rescaled times $t/t^* = 10^{-3}$, 10^{-2} , 10^{-1} , and 1 (bottom to top). Solid (dashed) lines show the space average (one standard deviation above and below average, as an estimate of the roughness). Data have been shifted vertically for visualization purposes.

and 1 in our simulations below to explore the effect of distance from the bifurcation point.

Phase profiles. To connect with KPZ physics, we describe the state of the oscillators along the ring by a set of phases $\{\phi_j(t)\}_{j=1}^L$, playing an analogous role to that of the local surface heights above a substrate, forming what we term a phase interface; see an example in Fig. 1 (inset) for the simple KS model. The dynamics of limit-cycle oscillators can also be described by the evolution of a single phase variable [25]. Moreover, close enough to the bifurcation point, one can perturbatively approximate such systems purely by phase dynamics via phase-reduction techniques [31, 45]. For SL oscillators, the uncoupled ($K = 0$) noiseless dynamics is characterized by the stable limit-cycle solution $z_j(t) = \sqrt{\mu} \exp i\theta(t)$, the (unwrapped) geometric phase $\theta_j(t) = t + \theta_j(0)$ [46] acting as a valid dynamical phase variable [25]. Off the limit cycle, an asymptotic phase [31] can be explicitly defined as $\phi_j(t) = \phi(z_j(t)) = \theta_j(t)$, i.e., the dynamic phase is the geometric phase everywhere in phase space (again considering the unwrapping of 2π oscillations) [47]. Thus, the instantaneous phases $\{\phi_j(t)\}_{j=1}^L$ are readily extracted from the states of the SL oscillators $\{z_j(t)\}_{j=1}^L$, on or off the limit cycle. A similar behavior is expected for vdP oscillators with $\mu \ll 1$. However, as μ is increased, vdP oscillators behave progressively as relaxation oscillators, moving at different speeds in different regions of noncircular limit cycles [44]. We thus define the phases $\{\phi_j(t)\}_{j=1}^L$ differently, namely, by

the crossing times of a Poincaré section [25]; see the Supplemental Material (SM) for details [48].

Scale-invariant roughening. The deviations of the local phases around their mean are measured by the roughness,

$$W(t) \equiv \langle [\phi(x, t) - \bar{\phi}(x, t)]^2 \rangle^{1/2}, \quad (2)$$

with the overbar denoting spatial average and the brackets denoting average over different noise realizations [13]. As we focus on large system sizes $L \gg 1$, we employ a continuum notation for the phase field, $\phi(x, t)$. The roughness saturates if the phase differences remain bounded in time, which only happens when all oscillators reach a common effective frequency. Thus, W reaching a steady-state value manifests synchronization (i.e., frequency entrainment) [39,40], which occurs for coupling $K > K_c$ above a synchronization threshold K_c .

Under the standard kinetic-roughening conditions [13–15] which extend equilibrium critical dynamics [1] to a host of other systems, local degrees of freedom are statistically correlated for distances below a time-increasing correlation length $\xi(t) \sim t^{1/z}$, where z is the dynamic exponent. After a saturation time $t^* \sim L^z$, $\xi(t)$ is comparable to L , the roughness saturating to a steady-state value $W^* \sim L^\alpha$, where the roughness exponent α characterizes the fractality of the profile [13,49]. Prior to that, in the growth regime, the roughness increases with time as $W(t) \sim t^\beta$, where the growth exponent $\beta = \alpha/z$. This behavior is indeed found in our phase profiles, as seen in Fig. 1 (main panel), which shows the roughness as a function of time rescaled by saturation values, $W(t/t^*)/W^*$, for rings of $L = 1000$ oscillators for the KS model with $\delta = 0$ and $\delta = \text{atan}5$ (see SM for justification of this choice [48]), and for the SL and the vdP oscillators. Here t^* is estimated as the time when W appears to reach the steady state, and W^* as the time average of W restricted to $t \geq t^*$. For the KS model with $\delta = 0$, the growth exponent follows closely that of the EW universality class, $\beta_{EW} = 1/4$, while in any other case it appears to be that of the KPZ universality class, $\beta_{KPZ} = 1/3$, with some deviations for vdP oscillators at small scales. While these results are robust against variations of the coupling strength as long as there is synchronization, for couplings below the synchronization threshold, $K < K_c$, the resulting (nonsynchronous) process appears to behave as random deposition in all cases, as expected for the KS model with $K = 0$ (see the SM for details [48]).

Correlations. Beyond global quantities like W , kinetic roughening also reveals itself in local correlation functions like the phase-difference correlation,

$$G(r, t) \equiv \langle [\phi(x+r, t) - \phi(x, t)]^2 \rangle, \quad (3)$$

which follows the Family-Vicsek (FV) dynamic scaling ansatz [13–15,50],

$$G(\ell, t) \sim \begin{cases} t^{2\beta}, & \text{if } t^{1/z} \ll \ell, \\ \ell^{2\alpha}, & \text{if } \ell \ll t^{1/z} \end{cases} = t^{2\beta} g(\ell/\xi(t)), \quad (4)$$

where $\ell \equiv |r| < L$, $g(y) \sim y^{2\alpha}$ for $y \ll 1$, and $g(y) \sim \text{cst.}$ for $y \gg 1$. The scaling form, Eq. (4), reflects the same growth and saturation of $W(t)$, restricted to a region of linear size ℓ [51]. In Fig. 2 (inset), we show $G(\ell, t)$ for the same systems considered in Fig. 1. The validity of the FV ansatz is

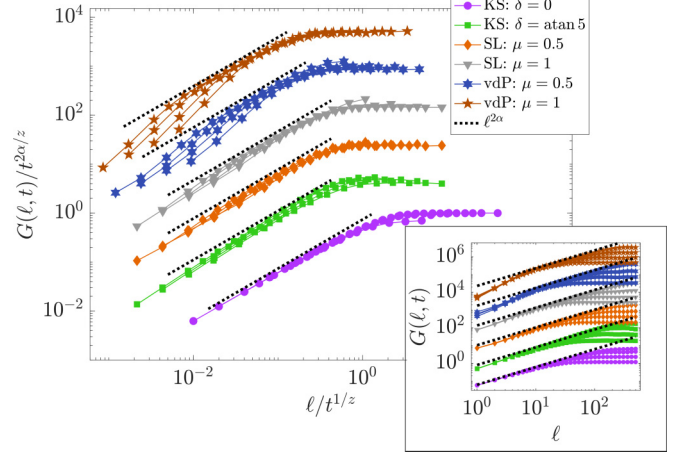


FIG. 2. Phase-difference correlation functions of rings of self-sustained oscillators. Main panel: Collapse as in Eq. (4) of the data displayed in inset for the phase-difference correlations $G(\ell, t)$, Eq. (3), for rings of KS, SL, and vdP oscillators, with the same parameters as in Fig. 1 (see legend). Exponent values are $\alpha = \alpha_{KPZ} = \alpha_{EW} = 1/2$ (dashed lines represent $\ell^{2\alpha} = \ell$), in all cases, and $z = z_{KPZ} = 3/2$, except for the KS ring with $\delta = 0$, for which $z = z_{EW} = 2$. Inset: Phase-difference correlation function $G(\ell, t)$, Eq. (3), at rescaled times $t/t^* = 1/8^z, 1/4^z, 1/2^z$, and 1 (bottom to top). An arbitrary vertical shift has been used for visualization purposes in both panels. Results based on 5000 realizations.

assessed in the main panel of Fig. 2, where it yields excellent data collapse, again except for vdP oscillators at small scales. Consistent with Fig. 1, the exponent values are 1D KPZ for all cases, $\alpha_{KPZ} = 1/2$ and $z_{KPZ} = 3/2$, except for the KS model with $\delta = 0$, where EW scaling holds, i.e., $\alpha_{EW} = 1/2$ and $z_{EW} = 2$. Further evidence on exponent values comes from Fourier-space correlations measured by the structure factor [51,52]; see the SM [48].

For 1D KPZ, two-point correlations actually take a universal form, known for the phase covariance [16,22]

$$C(r, t) \equiv \langle \phi(x, t) \phi(x+r, t) \rangle - \langle \bar{\phi}(t) \rangle^2, \quad (5)$$

which is shown in Fig. 3(a) for the same systems and parameters. In the growth regime [16,22], $C(\ell, t) = a_1 t^{2\beta} C(a_2 \ell/t^{1/z})$, where C is a scaling function, and a_1 and a_2 are constants [53]. The covariance of the Airy₁ process, expected for 1D KPZ and EW with PBCs [16,54], indeed occurs for all our systems, including the KS model with $\delta = 0$. The covariance of the Larkin model [55], also shown, applies to the same oscillator systems, but in the presence of columnar disorder [39,40].

Fluctuation statistics. A final remarkable trait of 1D KPZ is the universality of the fluctuations [14,16,22]. These are here studied in terms of rescaled phases,

$$\varphi_j \equiv \frac{\delta\phi_j(t_0 + \Delta t) - \delta\phi_j(t_0)}{(\Delta t)^\beta}, \quad (6)$$

where $\delta\phi_j(t) = \phi_j(t) - \bar{\phi}(t)$, t_0 is a reference time beyond the initial transient dynamics, and $t_0 + \Delta t$ lies within the growth regime. Division by $(\Delta t)^\beta$ removes the systematic time increase of fluctuations so, remarkably, the PDF of φ_j reaches a universal, time-independent form [14,16,22]. This is Gaussian for EW and the TW PDFs associated with the Gaussian

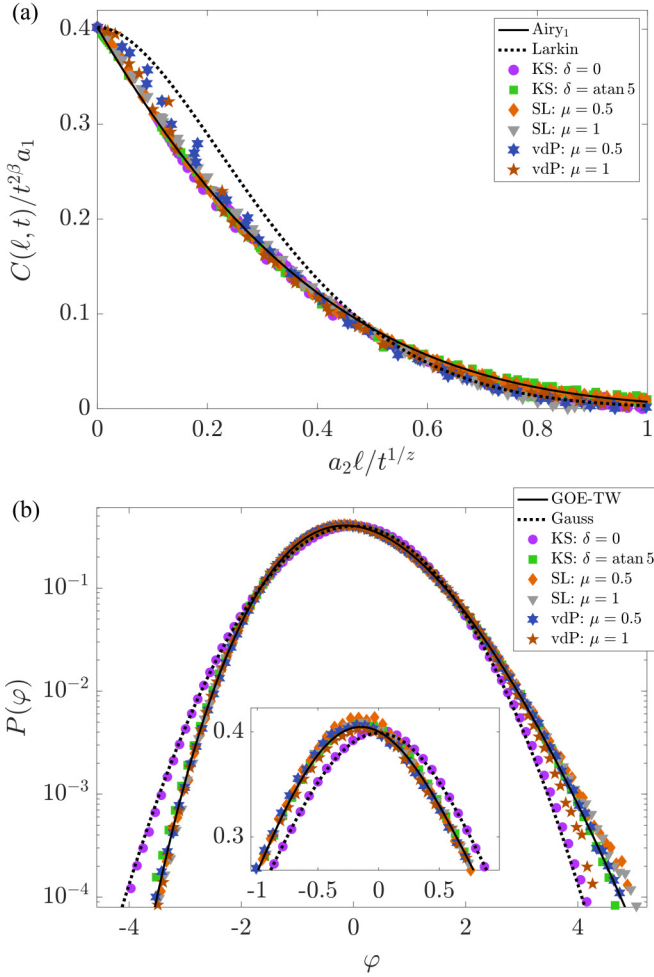


FIG. 3. Phase covariance and phase fluctuations of rings of self-sustained oscillators. (a) Rescaled phase covariance, Eq. (5), for rings of $L = 10\,000$ KS, SL, and vdP oscillators with parameters as in Fig. 1. The solid (dotted) line represents the covariance of the Airy_1 process (Larkin model). (b) Histogram of phase fluctuations, Eq. (6), in semilogarithmic scale for same rings of oscillators (see legend). Inset: Same data in linear scale, centered around the peak of the distribution. The solid (dotted) line represents a GOE-TW (Gaussian) PDF. Results based on 1000 realizations, with time points $t = t_0 + \Delta t$ equispaced along the growth regime, and t_0 at its initial stages. Exponent values are as in Fig. 2.

Orthogonal Ensemble (GOE-TW) for the KPZ equation with PBCs. Figure 3(b) shows the histogram of fluctuations defined as in Eq. (6) in semilogarithmic scale (main panel) and linear scale focusing around the distribution peak (inset). Again our numerics provide strong evidence that rings of KS oscillators with $\delta = 0$ are in the 1D EW universality class, while the rest of systems under considerations are in the 1D KPZ universality class. The only systematic deviations appear in the right tail for vdP oscillators sufficiently far from bifurcation, $\mu = 1$, at probabilities smaller than 10^{-3} , which might be related to the difficulty inherent in the definition of a dynamical phase for such systems.

Continuum approximation: Phase interface. We can rationalize the above findings by considering a relatively slow spatial variation of the phases, as for $K \gg K_c$, for a ring of oscillators with lattice constant a . A continuum approximation [39] which neglects terms $o(a^2)$ yields

$$\partial_t \phi(x, t) = \nu \partial_x^2 \phi(x, t) + \frac{\lambda}{2} [\partial_x \phi(x, t)]^2 + \text{noise}, \quad (7)$$

where $\nu \equiv a^2 \Gamma_\phi^{(1)}(0)$, $\lambda/2 \equiv a^2 \Gamma_\phi^{(2)}(0)$, $\Gamma_\phi(\Delta\phi)$ is the coupling function within phase reduction [40,45], and $\Gamma_\phi^{(k)}(0)$ its k th derivative at $\Delta\phi = 0$. For time-dependent noise, Eq. (7) is the 1D KPZ equation, where ν represents surface tension and λ growth along the local normal direction [13–15]. Detailed expressions of ν and λ are provided at the SM [48] for the oscillator systems we study. In particular, for the KS model we obtain $|\lambda/2\nu| = \tan \delta$ [48,56], which is zero [i.e., Eq. (7) becomes the EW equation] for $\delta = 0$ (Kuramoto model [29,31]), in conformity with our numerical results. Concerning both SL and vdP oscillators, we find [48] $|\lambda/2\nu| = \gamma \neq 0$, suggesting KPZ behavior as we numerically obtain [57–59].

Conclusions. A numerical study of rings of self-sustained oscillators with time-dependent noise reveals that their dynamical evolution into synchronization features GSI with all the traits of the 1D KPZ universality class, including TW statistics. A special case is the KS model with $\delta = 0$ (the celebrated Kuramoto model), which displays EW behavior. These conclusions agree with analytical expectations from a continuum approximation for large coupling K . The behavior of the Kuramoto model seems crucially related to the odd symmetry of its coupling function [60], leading to an up-down symmetry violated in the other cases [39]. The critical dynamics thus elucidated opens up questions [1,13], to be addressed for synchronization in general, concerning critical dimensions, (ir)relevant interactions, the role of symmetries, etc. Moreover, since for columnar disorder, synchronization generically displays the critical exponents and anomalous scaling ansatz of the columnar KPZ equation (the columnar EW equation for the KS model with $\delta = 0$), with a TW PDF and a Larkin covariance [39,40], intriguing questions arise on the interplay between this type of disorder and time-dependent noise.

As synchronization in 1D oscillator systems affected by time-dependent noise appears to display robust critical features generically in the KPZ class, their observation in experiment (on, e.g., electronic circuits, oscillating chemical reactions, or electrochemically active neurons or heart cells [25–27]) remains an alluring possibility.

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